MILITARY SPECIFICATION

FILTER, CRYSTAL

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions, and the preparation for delivery criterion for one type of Crystal Filter used on the Atomic Demolition Charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-S-60902(MU) - Soldering Electrical Connections for Nuclear Weapons Systems, Procedure for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810 - Environmental Test Methods
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169(MU) - Packaging, Packing and Marking for Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8808362 - Packaging and Marking for Storage and Shipments
8823154 - Filter, Crystal
8854464 - Electrical Test Panel
8854465 - Operating Instructions
8883539 - Bag, Packing for Repair Parts
ET-8823154 - Equipment Tabulation

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(Copies of specifications, standards, drawings, and publication required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The filter shall comply with all requirements specified on Drawing (dwg.) 8823154, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings.

3.4.1 Unclassified items. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.4.2 Classified items. Production quantities shall not be revealed. Random serial numbers shall be selected by the contractor in a manner which has been approved by the procuring agency. These numbers shall not be duplicated.

3.5 Environmental testing. The items shall comply with 4.3.3.1 before, during and after each environmental test when tested as specified in 4.4.1. Tests shall be conducted in the order specified.

3.5.1 Low temperature. The item shall comply with MIL-STD-810, method 502. Storage temperature shall be -65 degrees Fahrenheit (^{o}F). Operating temperature shall be -65°F.

3.5.2 High temperature. The item shall comply with MIL-STD-810, method 501, Procedure I. Operating temperature shall be 160°F.

3.5.3 Altitude. The item shall comply with MIL-STD-810, method 500, Procedure II. Pressure shall be equivalent to 30,000 ft. altitude.

3.5.4 Temperature shock. The item shall comply with MIL-STD-810, method 503.

3.5.5 Shock. The item shall comply with MIL-STD-810, method 516, Procedure I. Shock pulse shall be Fig. 516.2, Peak Value 15gs, duration 11 milliseconds.

3.5.6 Vibration. The item shall comply with MIL-STD-810, method 514, Procedure X. Test curve shall be in accordance with Fig. 6, curve AB. The equipment shall not be operated during the test.

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3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.6.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg. 8823154) for each dash number manufactured for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table I. Tests shall be conducted in order specified in table I. Contractor may perform tests before submission.

Table I

First article inspections and tests

Examinations and Tests
1. Classification of Characteristics (see 4.3.2.1)
2. Low Temperature (see 3.5.1)
3. High Temperature (see 3.5.2)
4. Altitude (see 3.5.3)
5. Temperature Shock (see 3.5.4)
6. Shock (see 3.5.5)
7. Vibration (see 3.5.6)

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1 1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Major characteristic table II

Lot size	Sample size
1- 155	100% inspection
156- 200	155
201- 250	175
251- 300	190
301- 350	200
351- 400	210
401- 450	220
451- 500	225
501- 550	230
551- 600	235
601- 650	240
651- 700	243
701- 750	247
751- 800	250
801- 900	254
901-1000	258
1001-1100	261
1101-1200	264
1201-1300	266
1301-1400	268
1401-1500	270
1501-1600	272

Minor characteristic table III

Lot size	Sample size
1- 25	100% inspection
26- 30	25
31- 35	26
36- 40	27
41- 45	28
46- 50	29
51- 55	30
56- 60	31
61- 65	32
66- 70	33
71- 75	34
76- 85	35
86- 95	36
96- 110	37
111- 130	38
131- 156	39
157- 190	40
191- 240	41
241- 340	42
341- 520	43
521-1120	44
1121 and over	45

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Filter, crystal (see dwg. 8823154).

Categories	Characteristics	Method of inspection
Critical:		
1.	Soldering in accordance with MIL-S-60902(MU)	Visual/Manual
Major:		
101.	Length	*Measure
102.	Location of pin and terminal holes	Gage
103.	Diameter of pin holes	Gage
104.	Depth of pin holes, min	Gage
105.	Height	*Measure
106.	Width	*Measure
107.	Marking	Visual

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108. Objective evidence (see 6.2) available for terminals in accordance with dwg. 8823154.....Visual
109. Objective evidence (see 6.2) available for finish in accordance with dwg. 8823154.....Visual
110. Material (see 3.1).....Visual
111. Proper sealing of unit.....Visual
112. Workmanship (see 3.6).....Visual/Manual

*Measure with suitable scale.

Minor: None defined.

4.3.2.1.2 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Packaging, packing and marking in accordance with MIL-STD-1169(MU).....Visual

Minor: None defined.

4.3.2.1.3 Bag, Packing, for Repair Parts (see dwg. 8883539 covering dwg. part number 8883539-4 and section 5).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Material (see 3.1).....Visual
102. Length before packing, min.....*Measure
103. Width before packing, min.....*Measure
104. Objective evidence (see 6.2) available for heat seal test in accordance with MIL-P-116.....Visual
105. Properly packed.....Visual
106. No tear, split, or crack in bag.....Visual

*Measure with suitable scale.

Minor:
201. Marking.....Visual
202. Workmanship (see 3.6).....Visual/Manual

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4.3.2.1.4 Packing and Marking for Storage and Shipments (see dwg. 8808362 and section 5).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Gross load weight, max	*Measure
102.	Contents tight in package	Manual
103.	Correctly packed	Visual/Manual
104.	Package material (see 3.11)	Visual

*Determine weight with a suitable scale.

Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.6)	Visual/Manual

4.3.3 Testing. All the tests specified in this paragraph shall be performed. Tests do not need to be performed in the order specified.

	Test	Test Classification
a.	Insertion loss	Major
b.	Response	Major
c.	Interchangeability	Major
d.	Downward force	Major
e.	High temperature	Major
f.	Low temperature	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Insertion loss, response, interchangeability, downward force. Each filter shall be tested for each drawing characteristic listed in accordance with 4.4.2. If any filter fails to comply with the drawing requirement, the filter shall be rejected and removed from the lot.

4.3.3.2 High temperature (see 3.5.2) and low temperature (see 3.5.1). Sample filters shall be selected at random from each inspection lot as specified in table IV and tested as specified in 4.4.1. If any filter fails to comply with 3.5.1 and 3.5.2 the inspection lot shall be rejected.

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Table IV

Lot Size	Sample Size for No Failures
1-25	100% inspection
26-30	25
31-35	26
36-40	27
41-45	28
46-50	29
51-55	30
56-60	31
61-65	32
66-70	33
71-75	34
76-85	35
86-95	36
96-110	37
111-130	38
131-156	39
157-190	40
191-240	41
241-340	42
341-520	43
521-1120	44
1121 and over	45

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8823154. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Environmental testing (see 3.5). The item shall be tested in accordance with MIL-STD-810 using the applicable method specified in the requirement.

4.4.2 Insertion loss, response, interchangeability and downward force (see 4.3.3.1). The filter shall be tested in accordance with dwg. 8823154.

5. PREPARATION FOR DELIVERY

5.1 Production filter crystals.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The filter shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The filter shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169 (MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The filter shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair part filter crystals.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The filter shall be packaged in accordance with dwgs. 8883539 and 808362 using the specified moisture barrier.

5.2.1.2 Level C. The filter shall be packaged in accordance with dwg. 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The filter shall be packed in accordance with dwg. 8883539 and 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs. 8883539 and 8808362, as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

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6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification includes the requirements of Purchase Description PA-PD-1890 (Rev 2) dated 21 August 1963.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1125-A021

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-F-50280(MU) Filter, Crystal

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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